GENERAL ASSEMBLY

INTRO TO DATA SCIENCE

LECTURE 4: DATA VISUALIZATION & EDA

Announcements

- **Meetups!**
  - **Data Driven NYC: @Bloomberg (10/14)**
  - **NY Python Meetup: Lightning talks (10/9)**
  - **NY Data Science: DataViz w/ Plotly (10/14)**
- **Solution to Lab posted**

AGENDA

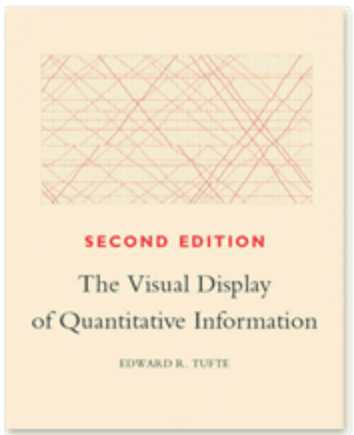
I. Data Viz: Story Telling

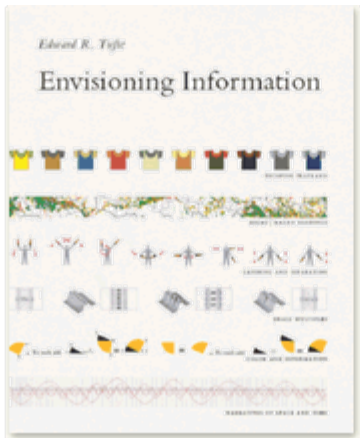
II. Data Viz: EDA

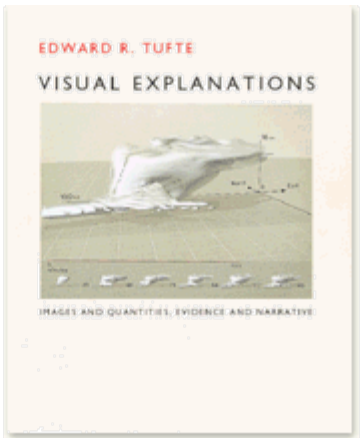
III. Python Data Viz Tools

IV. Lab: practice

I. DATA VISUALIZATION: STORY TELLING


SECOND EDITION
The Visual Display
of Quantitative Information
EDWARD R. TUFTE


Edward R. Tufte
Envisioning Information


EDWARD R. TUFTE
VISUAL EXPLANATIONS
IMAGES AND QUANTITIES, EVIDENCE AND NARRATIVE



Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

Dressée par M. Minard, Inspecteur Général des Ponts et Chaussées en retraite.

Paris, le 20 Novembre 1869.

Les nombres d'hommes présents sont représentés par les largeurs des zônes colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en travers des zônes. Le rouge désigne les hommes qui entrent en Russie, le noir ceux qui en sortent. — Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M.M. Thiers, de Ségur, de Fezensac, de Chambray et le journal inédit de Jacob, pharmacien de l'Armée depuis le 28 Octobre.

Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Jérôme et du Maréchal Davoust qui avaient été détachés sur Minsk et Mobilow et ont rejoint vers Orscha et Witebsk, avaient toujours marché avec l'armée.

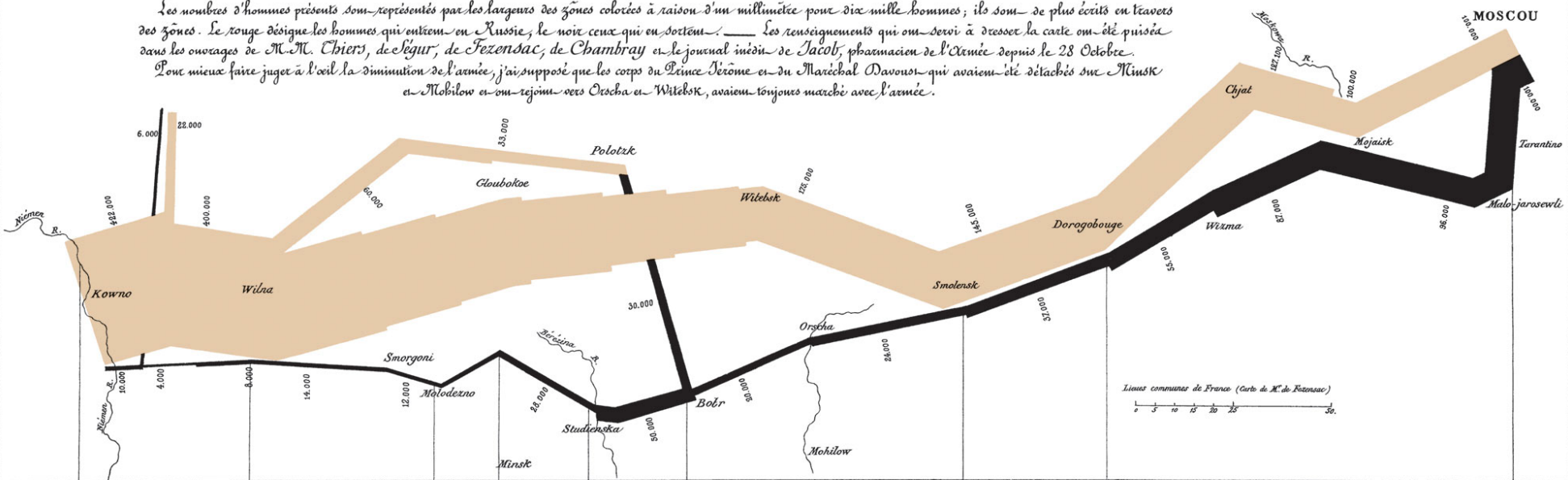
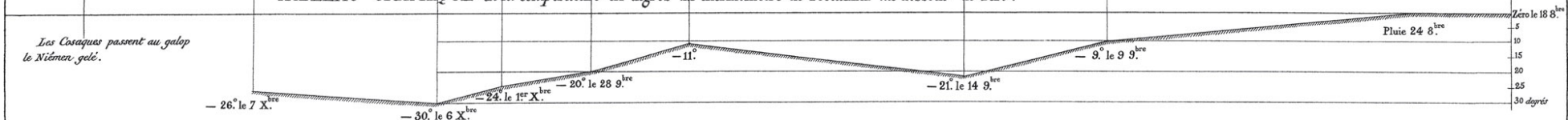
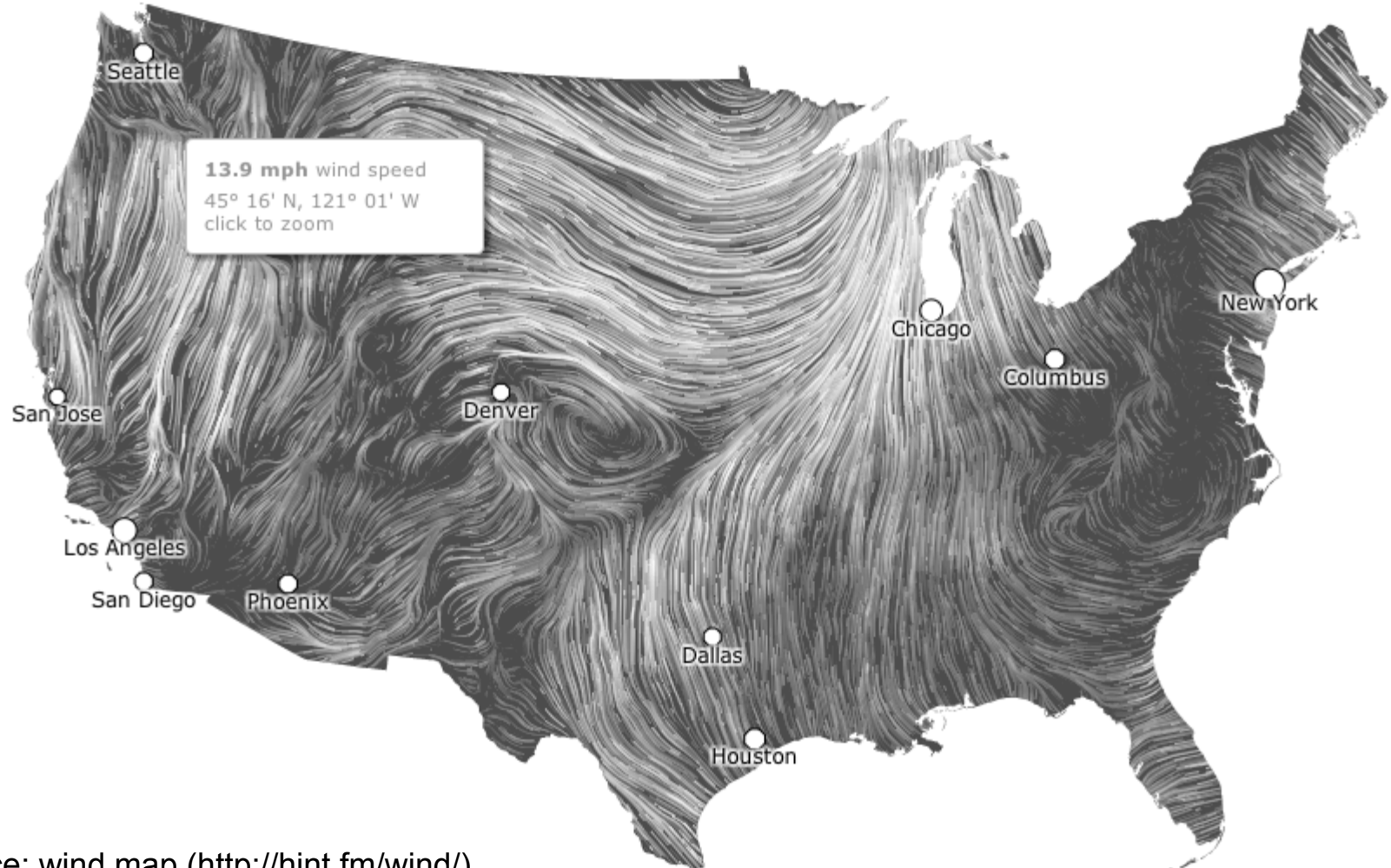


TABLEAU GRAPHIQUE de la température en degrés du thermomètre de Réaumur au dessous de zéro.

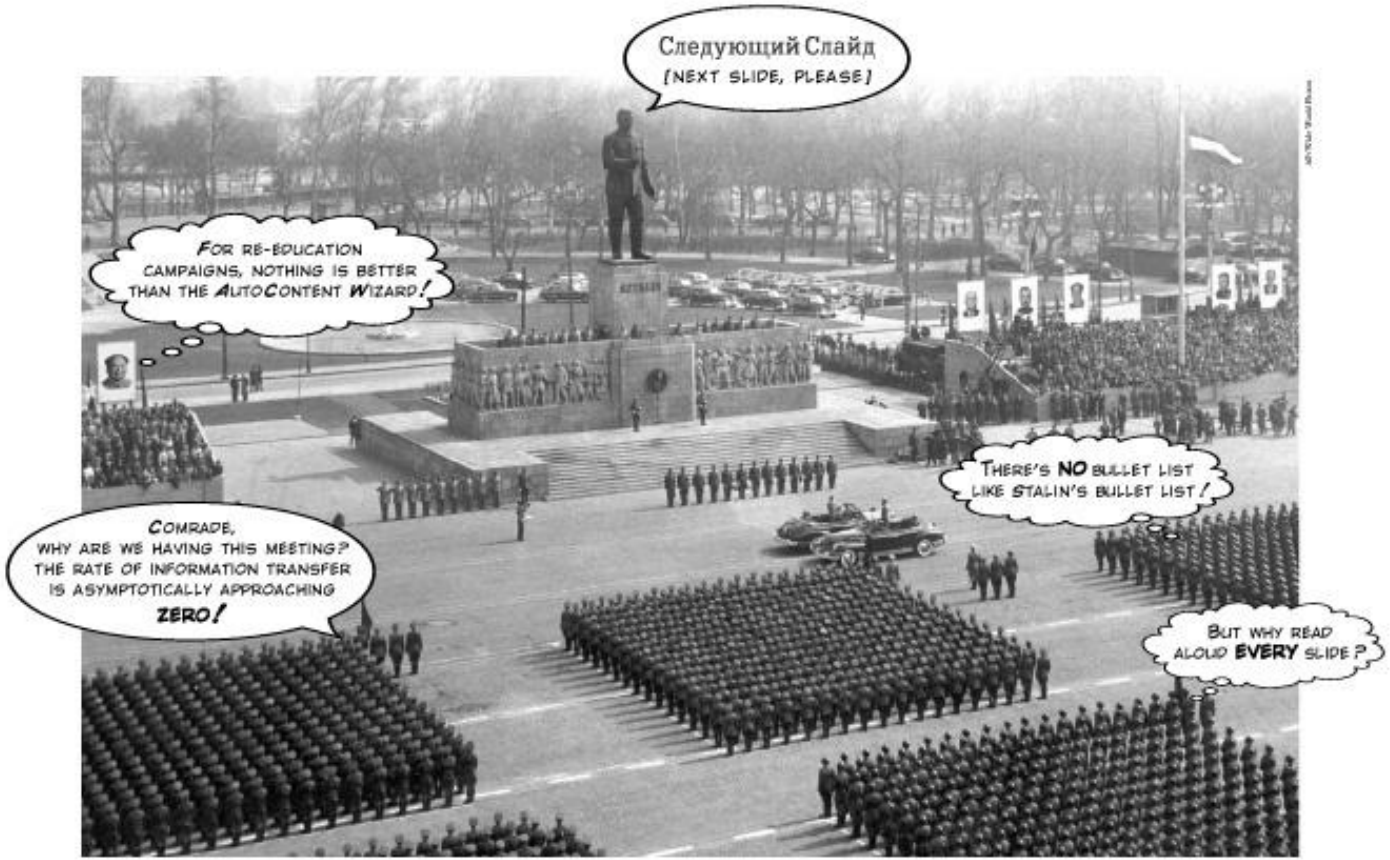


Autog. par Regnier, 8. Pas. Ste Marie St Gelin à Paris.

Imp. Lith. Regnier et Dourdet.

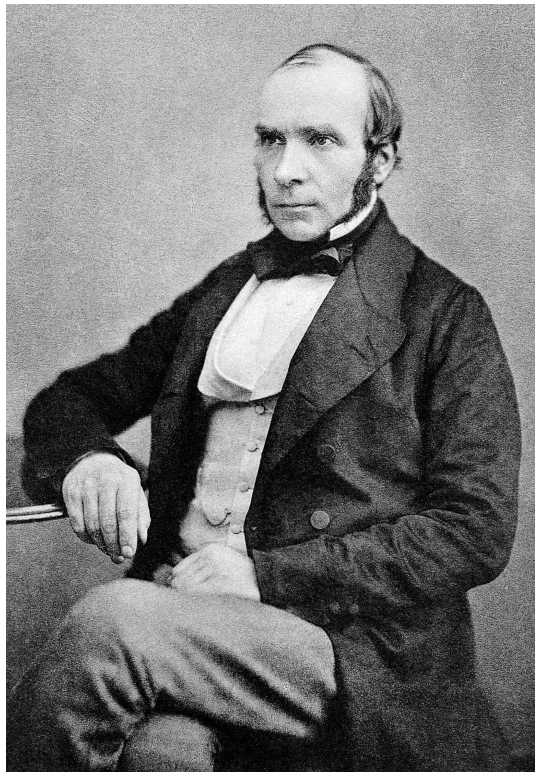


source: wind map (http://hint.fm/wind/)

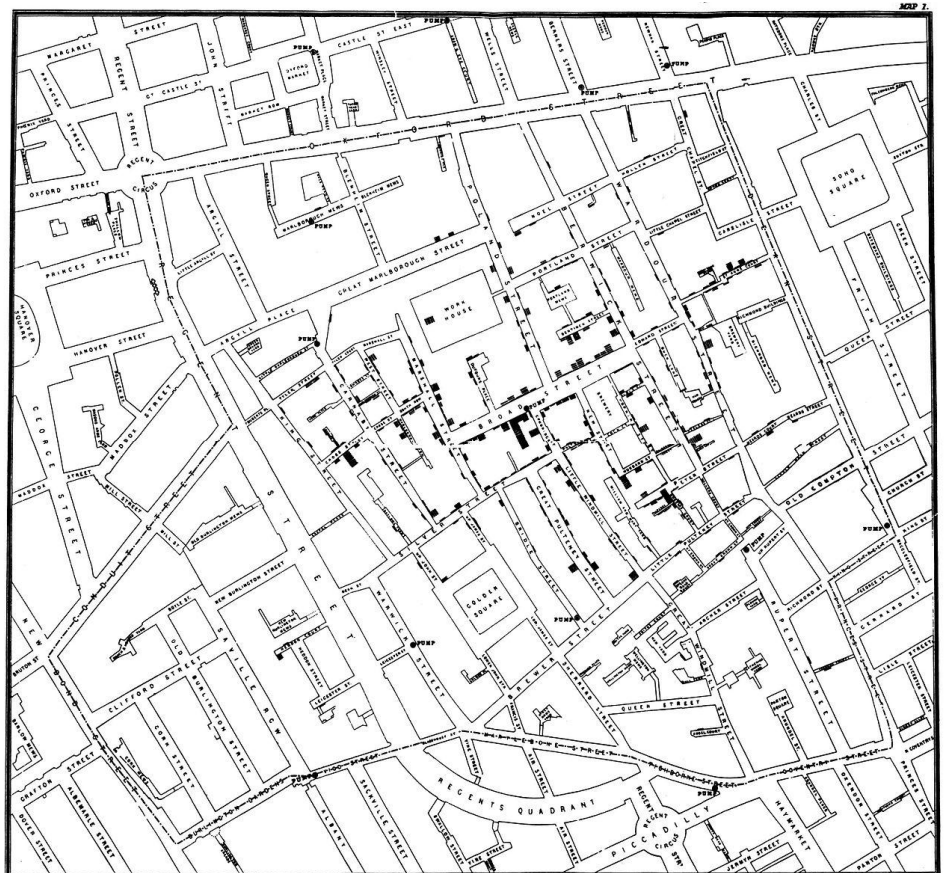


Edward Tufte, *The Cognitive Style of PowerPoint*


Summary
New nation
Civil war
Dedicate field
Dedicated to unfinished work
New birth of freedom
Government not perish
11/19/1863

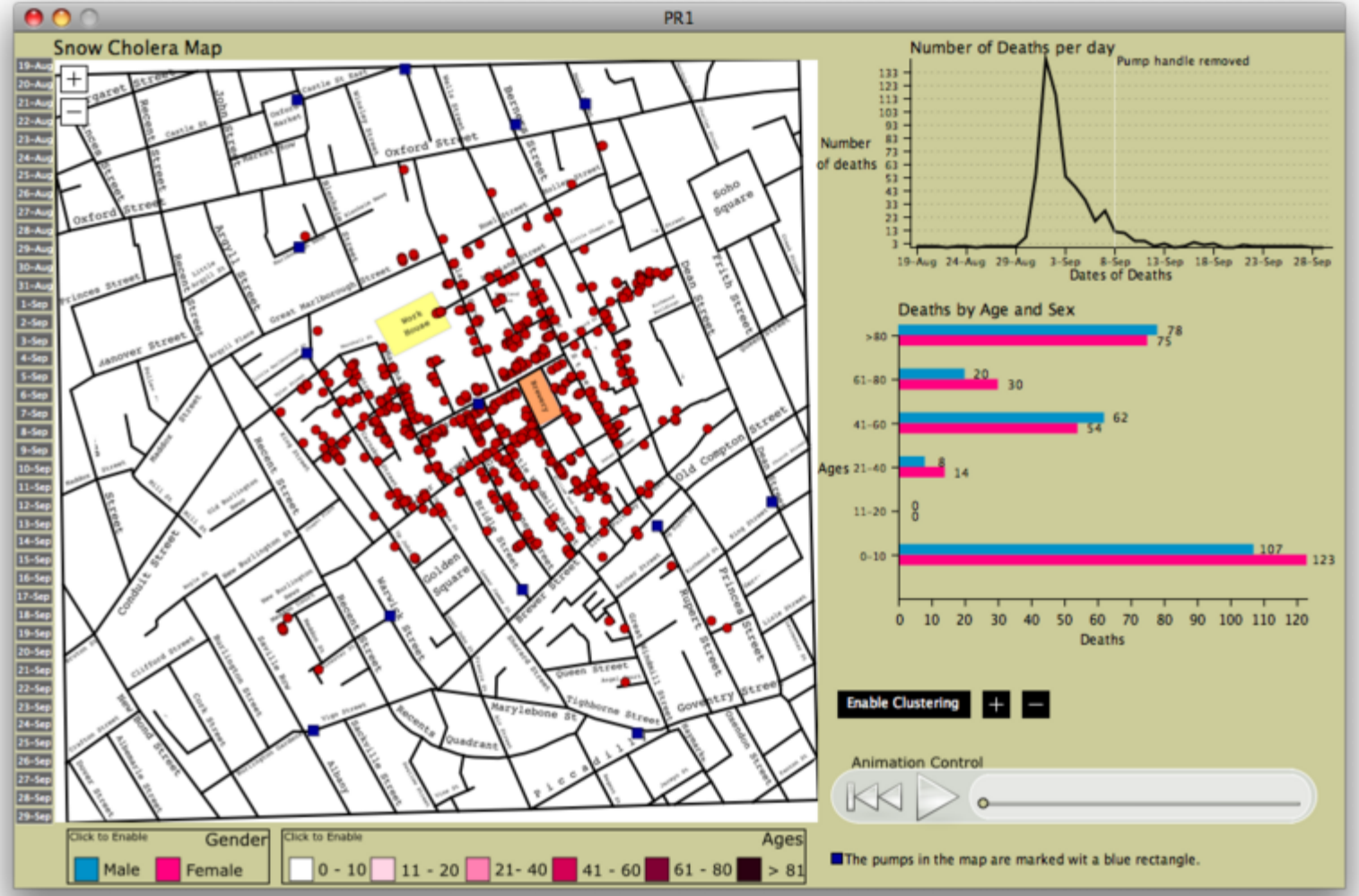


John Snow



C. F. Cheffins, Lith. Southampton St. London.

SCALE 30 INCHES TO A MILE.


PR1
Snow Cholera Map
Oxford Street
Soho Square
Princes Street
Hanover Street
Great Marlborough Street
Work House
Broadwick Street
Old Compton Street
Golden Square
Warwick Street
Regent Street
Brewer Street
Marylebone St
Piccadilli
Pighborne Street
Queen Street
Coventry Street
Number of Deaths per day
Pump handle removed
Number of deaths
Dates of Deaths
Deaths by Age and Sex
>80
61-80
41-60
Ages 21-40
11-20
0-10
78
75
20
30
62
54
8
14
0
0
107
123
Deaths
Enable Clustering
Animation Control
Click to Enable
Gender
Male
Female
Click to Enable
Ages
0 - 10
11 - 20
21- 40
41 - 60
61 - 80
> 81
The pumps in the map are marked wit a blue rectangle.

CRIMINAL ACTIVITY OF GOVERNMENT INFORMANTS

CRIME	CARDINALE	LOFARO	MALONEY	POLISI	SENATORE	FORONJY	CURRO
MURDER	X	X					
ATTEMPTED MURDER		X	X				
HEROIN POSSESSION AND SALE	X	X		X			X
COCAINE POSSESSION AND SALE	X		X	X			
MARIJUANA POSSESSION AND SALE							X
GAMBLING BUSINESS		X		X		X	
ARMED ROBBERIES	X		X	X	X		X
LOANSHARKING		X		X			
KIDNAPPING			X	X			
EXTORTION			X	X			
ASSAULT	X		X	X			X
POSSESSION OF DANGEROUS WEAPONS	X	X	X	X	X		X
PERJURY		X				X	
COUNTERFEITING					X	X	
BANK ROBBERY			X	X			
ARMED HIJACKING				X	X		
STOLEN FINANCIAL DOCUMENTS			X	X	X		
TAX EVASION				X		X	
BURGLARIES	X	X		X	X		
BRIBERY		X		X			
THEFT: AUTO, MONEY, OTHER			X	X	X	X	X
BAIL JUMPING AND ESCAPE			X	X			
INSURANCE FRAUDS					X	X	
FORGERIES				X	X		
PISTOL WHIPPING A PRIEST	X						
SEXUAL ASSAULT ON MINOR							X
RECKLESS ENDANGERMENT							X

II. Data Visualization: Exploratory Data Analysis

Exploratory Data Analysis

"Exploratory Data Analysis" is an attitude, a state of flexibility, a willingness to look for those things that we believe are not there, as well as those we believe to be there.

- John Tukey

Exploratory Data Analysis

News Flash:

EXPLORATORY DATA ANALYSIS

News Flash:

You've already been doing Exploratory Data Analysis

Exploratory Data Analysis: Goals

- Gain intuition about the data
- Make comparisons between distributions
- Sanity checking
- Find missing data, outliers
- Summarize the data/ summary statistics

Exploratory Data Analysis: Summary

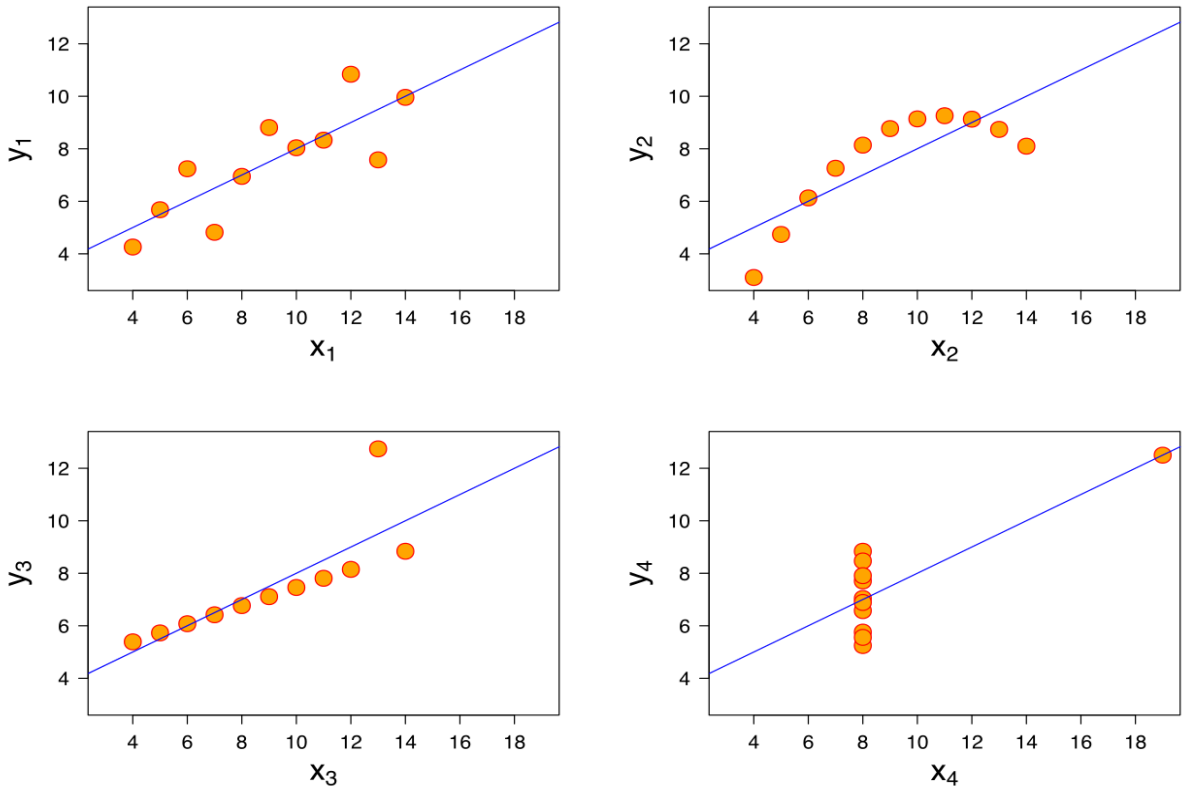
Statistics

- ***Mean*** (as one of the averages or measures of center)
- ***Variance*** (as covariance with self)
- ***Correlation*** (as measure of varying together linearly)

ANSCOMBE'S QUARTET

	I		II		III		IV	
	x	y	x	y	x	y	x	y
	10	8,04	10	9,14	10	7,46	8	6,58
	8	6,95	8	8,14	8	6,77	8	5,76
	13	7,58	13	8,74	13	12,74	8	7,71
	9	8,81	9	8,77	9	7,11	8	8,84
	11	8,33	11	9,26	11	7,81	8	8,47
	14	9,96	14	8,1	14	8,84	8	7,04
	6	7,24	6	6,13	6	6,08	8	5,25
	4	4,26	4	3,1	4	5,39	19	12,5
	12	10,84	12	9,13	12	8,15	8	5,56
	7	4,82	7	7,26	7	6,42	8	7,91
	5	5,68	5	4,74	5	5,73	8	6,89
SUM	99,00	82,51	99,00	82,51	99,00	82,50	99,00	82,51
AVG	9,00	7,50	9,00	7,50	9,00	7,50	9,00	7,50
STDEV	3,32	2,03	3,32	2,03	3,32	2,03	3,32	2,03

ANSCOMBE'S QUARTET



EXPLORATORY DATA ANALYSIS: PREPROCESSING

- dealing with categorical variables
- string manipulation
- data transformation:

Exploratory Data Analysis:

Transformations

1. Convenience
2. Reducing skewness
3. Equal spreads
4. Scaling/ Normalization

EXPLORATORY DATA ANALYSIS:

TRANSFORMATIONS

1. Convenience
   (e.g. percentages vs. original data, sines vs degrees)
2. Reducing skewness
   a. take roots or logarithms or reciprocals (common)
   b. take squares, cubes
3. Equal “spreads”
   a. Each data set or subset having about the same spread or variability is ***homoscedasticity***: the opposite is called ***heteroscedasticity***
4. Scaling/ Normalization

EXPLORATORY DATA ANALYSIS:

TRANSFORMATIONS

Most Common:

1. Reciprocal
    a. e.g. persons per doctor becomes doctors per person
2. Logarithm
3. Square/Cube root
4. Power

Exploratory Data Analysis: Goals

questions:
what am I optimizing for?
what are the interesting features of the dataset?

outcomes:
need more data
sense of scope of problem

Exploratory Data Analysis: IRL

EXPLORATORY DATA ANALYSIS: IRL

Celery Job Queue EDA:

- How can we detect job completion rates that are outside acceptable bounds?
- Are there any daily or weekly trends in jobs?
- Are there bottlenecks anywhere in job sequences?

EXPLORATORY DATA ANALYSIS: GET THE DATA

```
In [3]:
def report(connection, job, sent_date):
        session = connection.connect('monitoring')
        session.row_factory = dict_factory

        job_data = list(session.execute("SELECT * FROM jobs WHERE task=%s AND sent_date=%s", [job, sent_date]))

        frame = pd.DataFrame(job_data)
        frame['sent_date'] = sent_date
        frame['queued'] = frame['received'] - frame['sent']
        frame['pending'] = frame['started'] - frame['received']
        frame['runtime'] = frame['ended'] - frame['started']

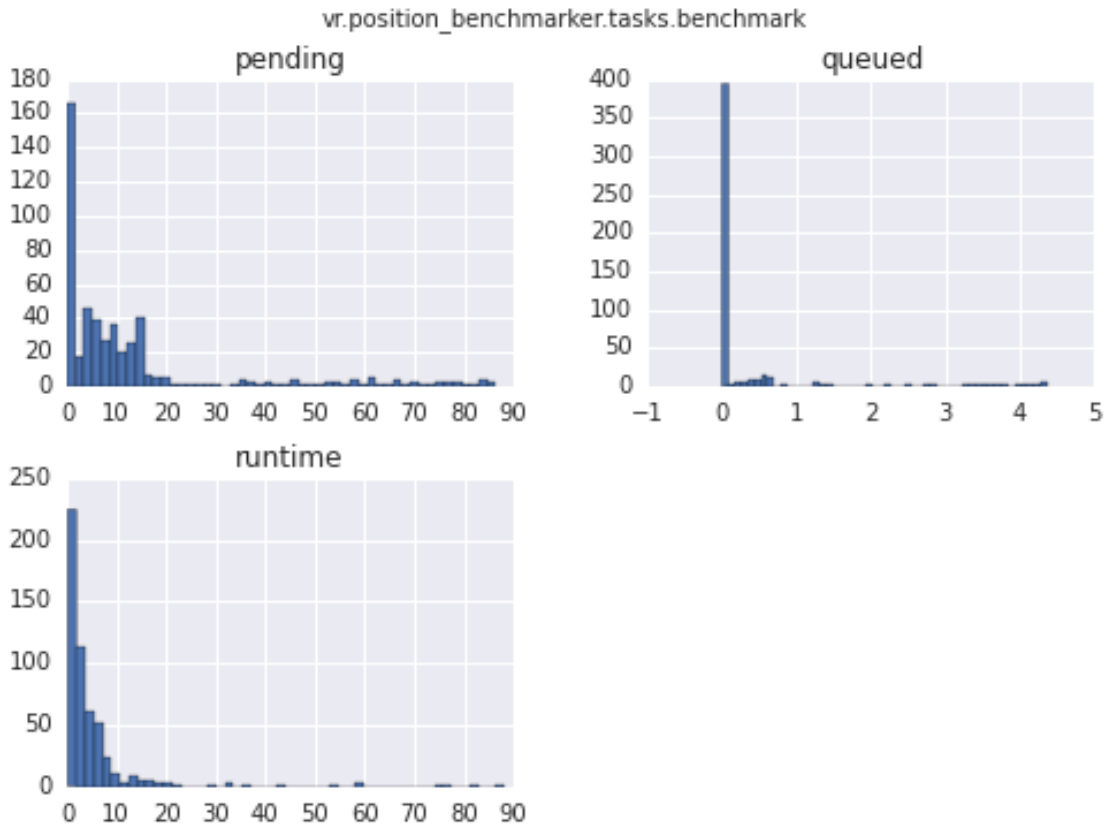
        frame['queued'] = frame['queued'].apply(lambda x: x.total_seconds())
        frame['pending'] = frame['pending'].apply(lambda x: x.total_seconds())
        frame['runtime'] = frame['runtime'].apply(lambda x: x.total_seconds())

        total = frame['task'].count()
        states = frame['state'].value_counts().to_dict()

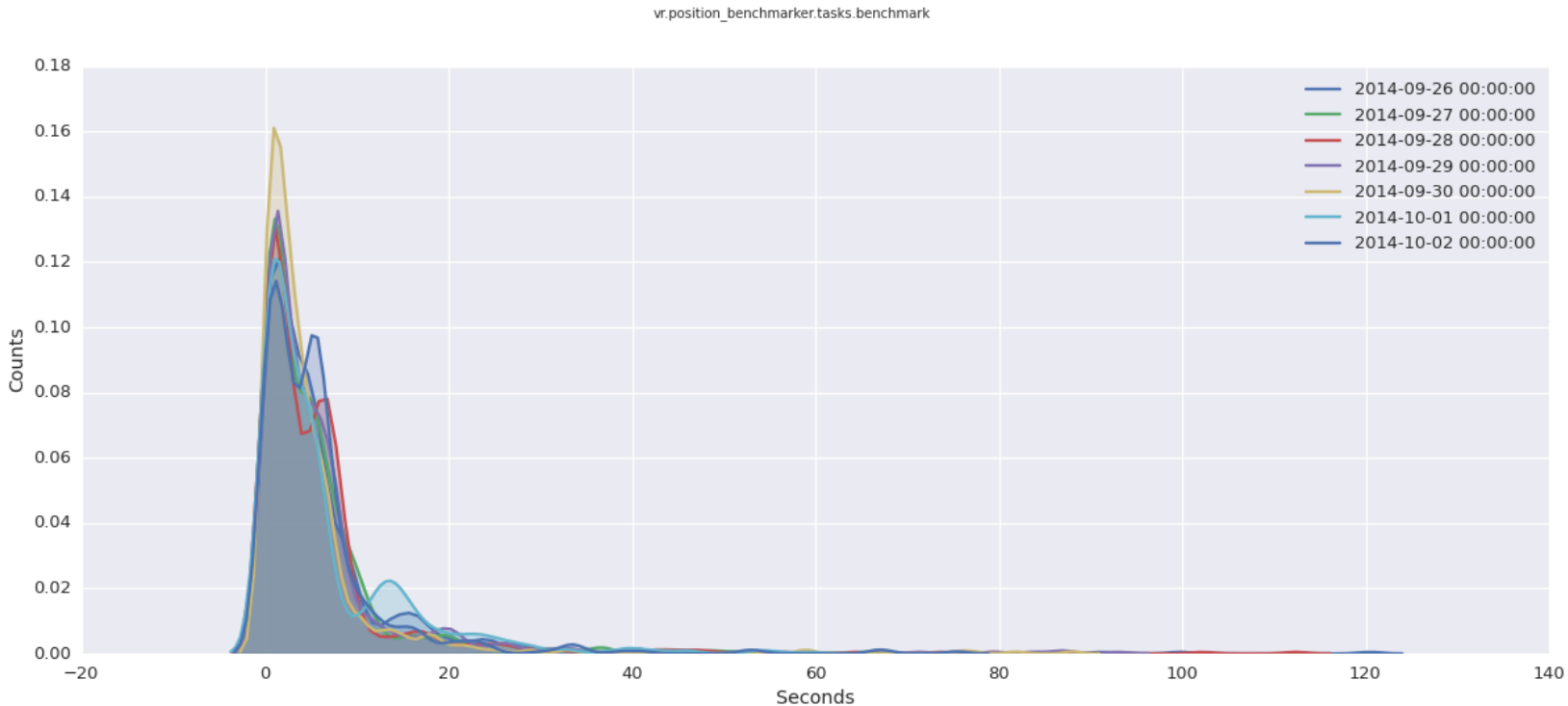
        print '{}\t{}'.format(job, sent_date)
        # all columns: [args, ended, error, fp, kwargs, np, received, sent, sent_date, started, state, task, uui
        # drop unused columns
        frame = frame.drop('args', 1)
        frame = frame.drop('ended', 1)
        frame = frame.drop('error', 1)
        frame = frame.drop('kwargs', 1)
        frame = frame.drop('received', 1)
        frame = frame.drop('sent', 1)
        frame = frame.drop('uuid', 1)
        frame = frame.drop('started', 1)

        return frame
```

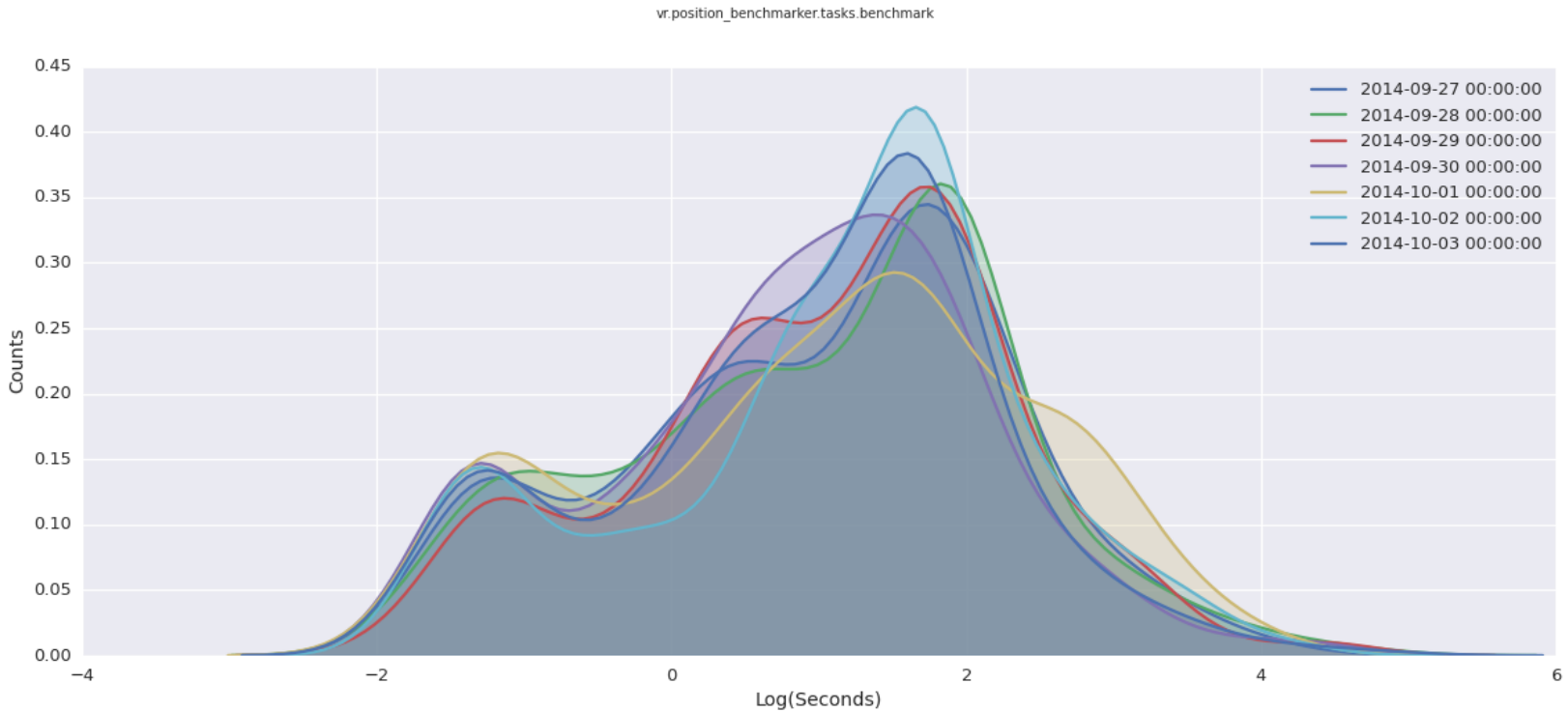
Exploratory Data Analysis: Visualize the data



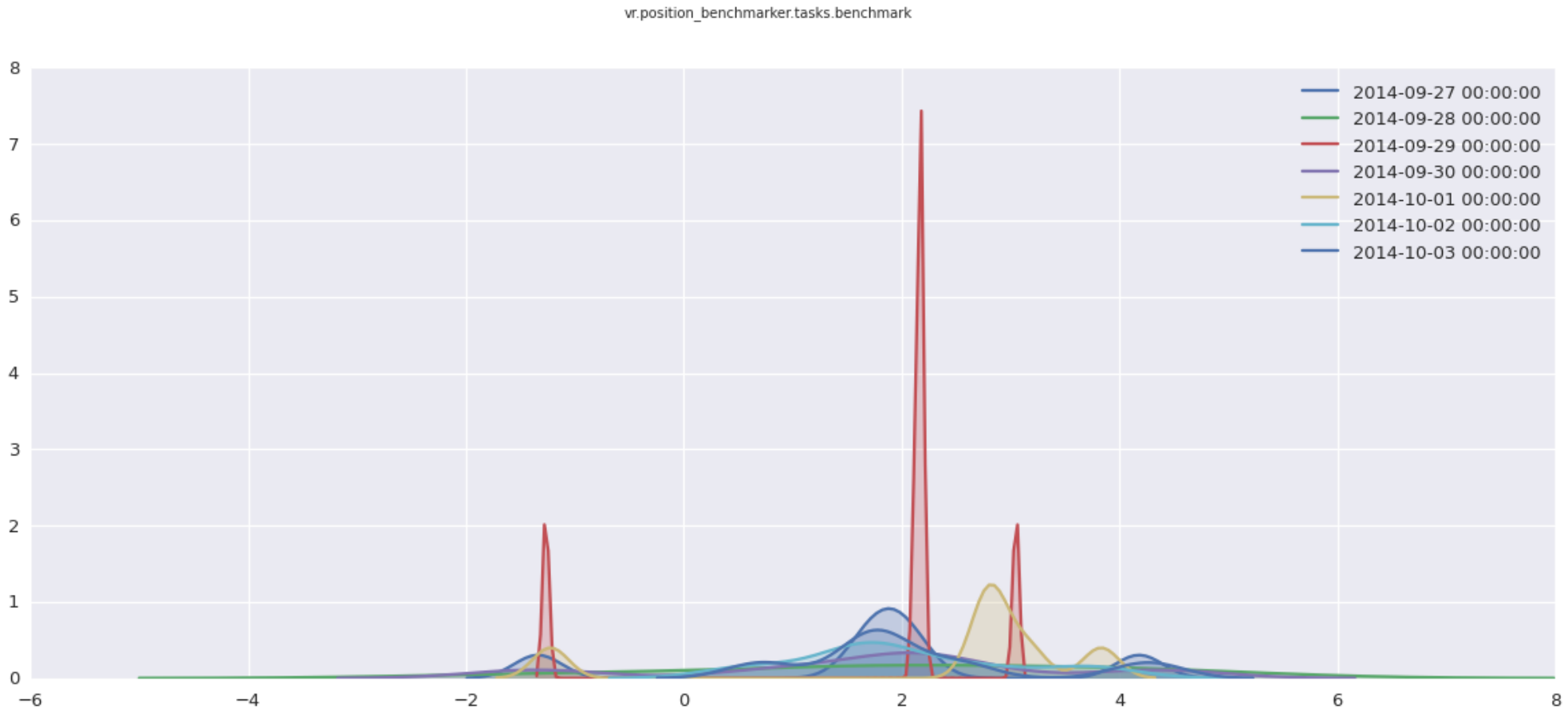
Exploratory Data Analysis: Visualize the Data



Exploratory Data Analysis: Transform the Data



Exploratory Data Analysis: Filter the



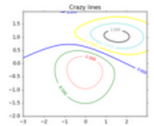
IV. LAB: PRACTICE

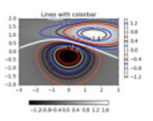
Exploratory Data Analysis: Tools

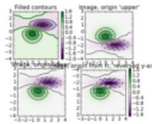
- matplotlib
  - seaborn
  - vincent
  - bokeh
- d3
  - rCharts
  - NVD3
- ggplot (Python port and R2Py)
- Plot.ly
- Gephi

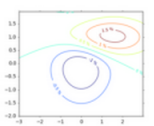
EXPLORATORY DATA ANALYSIS: *MATPLOTLIB*

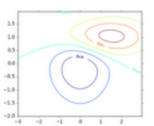
Use The [Gallery]

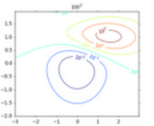

contour_demo

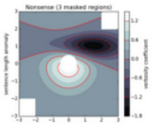

contour_demo

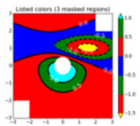

contour_image

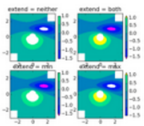

contour_label_demo

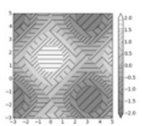

contour_label_demo

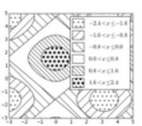

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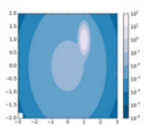

contourf_demo

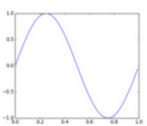

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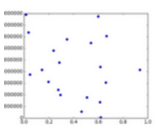

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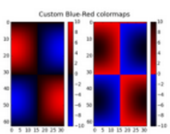

contourf_hatching

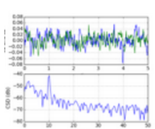

contourf_hatching

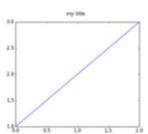

contourf_log

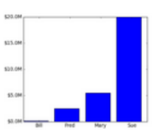

coords_demo

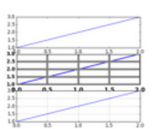

coords_report

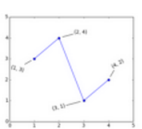

custom_cmap

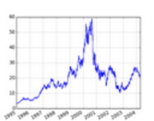

csd_demo

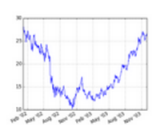

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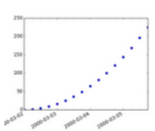

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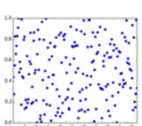

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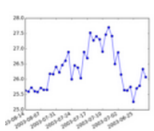

dashpointlabel

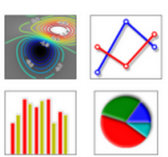

date_demo1


date_demo2


date_demo_convert


date_demo_rrule

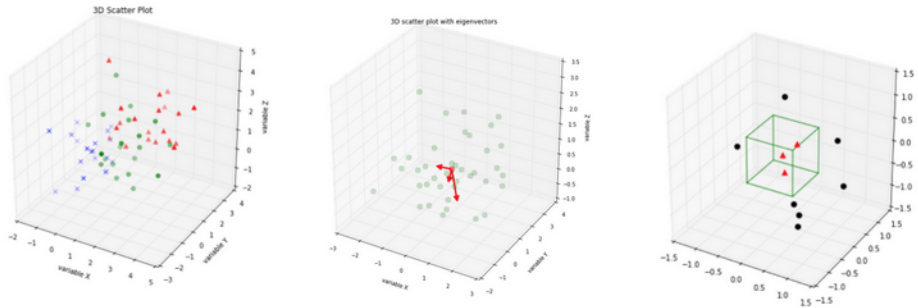

date_index_formatter

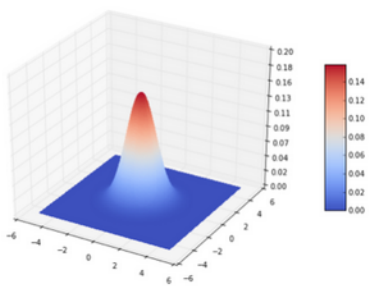

demo_agg_filter

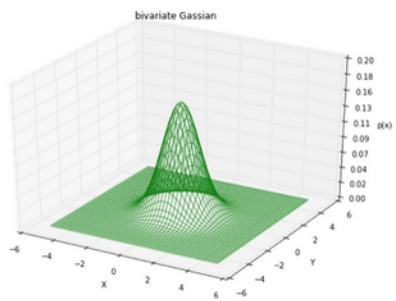

demo_annotation_box

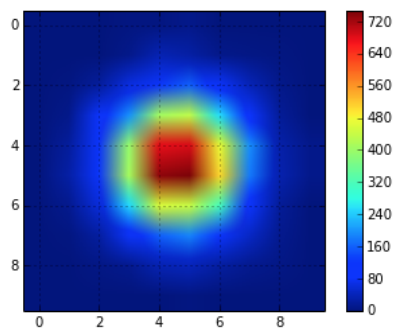
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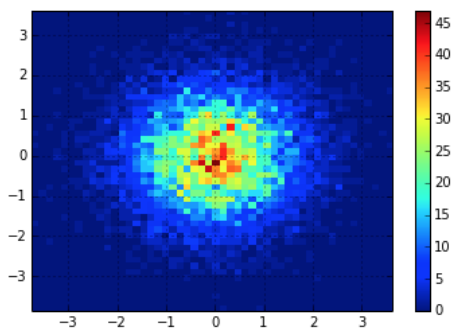
More examples:











EXPLORATORY DATA ANALYSIS: *MATPLOTLIB*

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- animation Examples
  - animate_decay, basic_example, basic_example_writer, bayes_update, double_pendulum_animated, dynamic_image
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  - agg_oo, barchart_demo, bbox_intersect, collections_demo, colorbar_only, compound_path
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  - legend_demo, line_with_text, logo2, mathtext_asarray, patch_collection, power_norm_demo
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  - scatter_piecharts, skewt, span_regions, two_scales, unicode_minus, watermark_image
  - watermark_text
- axes_grid Examples
  - demo_axes_divider, demo_axes_grid, demo_axes_grid2, demo_axes_hbox_divider, demo_axes_rgb, demo_axisline_style
  - demo_colorbar_with_inset_locator, demo_curvelinear_grid, demo_curvelinear_grid2, demo_edge_colorbar, demo_floating_axes, demo_floating_axis
  - demo_imagegrid_aspect, demo_parasite_axes2, inset_locator_demo, inset_locator_demo2, make_room_for_ylabel_using_axesgrid, parasite_simple2
  - scatter_hist, simple_anchored_artists, simple_axesgrid, simple_axesgrid2, simple_axisline4
- color Examples
  - color_cycle_demo, colormaps_reference, named_colors
- event_handling Examples
  - close_event, data_browser, figure_axes_enter_leave, idle_and_timeout, keypress_demo, lasso_demo
  - legend_picking, looking_glass, path_editor, pick_event_demo, pick_event_demo2, pipong
  - poly_editor, pong_gtk, resample, test_mouseclicks, timers, trifinder_event_demo
  - viewlims, zoom_window
- images_contours_and_fields Examples
  - image_demo, image_demo_clip_path, interpolation_methods, interpolation_none_vs_nearest, pcolormesh_levels, streamplot_demo_features
  - streamplot_demo_masking
- lines_bars_and_markers Examples
  - barh_demo, fill_demo, fill_demo_features, line_demo_dash_control, line_styles_reference, marker_fillstyle_reference
  - marker_reference, scatter_with_legend
- misc Examples
  - contour_manual, font_indexing, ftface_props, image_thumbnail, longshort, multiprocess
  - rasterization_demo, rc_traits, rec_groupby_demo, rec_join_demo, sample_data_demo, svg_filter_line
  - svg_filter_pie, tight_bbox_test
- mplot3d Examples
  - 2dcollections3d_demo, bars3d_demo, contour3d_demo, contour3d_demo2, contour3d_demo3, contourf3d_demo
  - contourf3d_demo2, hist3d_demo, lines3d_demo, lorenz_attractor, mixed_subplots_demo, offset_demo
  - pathpatch3d_demo, polys3d_demo, quiver3d_demo, rotate_axes3d_demo, scatter3d_demo, subplot3d_demo
  - surface3d_demo, surface3d_demo2, surface3d_demo3, surface3d_radial_demo, text3d_demo, tricontour3d_demo
  - tricontourf3d_demo, trisurf3d_demo, trisurf3d_demo2, wire3d_animation_demo, wire3d_demo
- pie_and_polar_charts Examples
  - pie_demo_features, polar_bar_demo, polar_scatter_demo
- pylab_examples Examples
  - accented_text, agg_buffer_to_array, alignment_test, anchored_artists, animation_demo, annotation_demo
  - annotation_demo2, annotation_demo3, anscombe, arctest, arrow_demo, arrow_simple_demo
  - aspect_loglog, axes_demo, axes_props, axes_zoom_effect, axhspan_demo, axis_equal_demo
  - bar_stacked, barb_demo, barchart_demo, barchart_demo2, barcode_demo, boxplot_demo
  - boxplot_demo2, boxplot_demo3, break, broken_axis, broken_barh, centered_ticklabels
  - clippedline, cohere_demo, color_by_yvalue, color_demo, colorbar_tick_labelling_demo, colours

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Explore whatever datasets you find most interesting:

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2. IMDB dataset
3. NYT CTR dataset
4. Africa Soil Challenge